[INCH-POUND]
A-A-59351
28 September 1999
SUPERSEDING
MIL-G-45072G
10 November 1988

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, DRILL

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-G-45072G for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for floor-mounted or bench-mounted grinding machines used to sharpen drill bits.

2. CLASSIFICATION. Floor-mounted machines shall be classified as type I and bench-mounted machines shall be classified as type II. Each type shall be further classified in various sizes which shall represent the drive motor horsepower ratings as listed in table I. The grinding machine type and size to be supplied shall be specified in the acquisition order (see 7.4).

TABLE I. Drill bit grinding machine classifications.

Type	Size	Horsepower
I (Floor mounted)	1	1/3
	2	1/2
	3	1/2
	4	1
II (Bench mounted)	1	1/4
	2	1/3
	3	1/2

3. SALIENT CHARACTERISTICS. The grinding machine shall comply with the requirements specified in this document and in American National Standards Institute (ANSI) standard B11.9, "Safety Requirements for the Construction, Care, and Use of Grinding Machines".

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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3.1 Dimensions. Type I grinding machines shall meet the dimensional requirements specified in table II. Type II machines shall meet the dimensional requirements specified in table III.

TABLE II. Floor-mounted grinding machine dimensional requirements.

Characteristics	Requirements			
	Size 1	Size 2	Size 3	Size 4
Drill diameter (inches)	1/16 to 3/4	1/8 to 1-1/8	1/8 to 2	1/4 to 3
Drill length (inches)	2 to 10	2 to 13	3-1/2 to 20	3-1/2 to 25
Point angle (degrees)	90 to 160	90 to 160	90 to 160	90 to 160
Lip relief angle (degrees)	6 to 15	6 to 15	6 to 15	6 to 15
Flutes (number)	2	2	2, 3, or 4	2, 3, or 4

TABLE III. Bench-mounted grinding machine dimensional requirements.

Characteristics	Requirements		
	Size 1	Size 2	Size 3
Drill diameter (inches)	1/16 to 1/2	1/16 to 3/4	1/8 to 1-1/8
Drill length (inches)	2 to 7	2 to 10	2 to 13
Point angle (degrees)	90 to 160	90 to 160	90 to 160
Lip-relief angle (degrees)	6 to 15	6 to 15	6 to 15
Flutes (number)	2	2	2

3.2 Tolerances. The grinding machine shall grind a drill bit point angle within ±5 degrees of the setting through the full machine range of 90 to 160 degrees. The lip-relief angles shall be equal within ±1 degree. The point-angle and lip-relief angles shall be concentric around the drill body cylinder with no irregularities of more than 0.002 inch Total Indicator Readout (T.I.R). The ground finish of each drill point shall have no surface irregularities of more than 20 micro-inches.

3.3 Components. The grinding machine components shall include a grinding wheel, drive motor, and feed mechanism. The wheel shall comply with ANSI specifications B74.2, "Specifications for Shapes and Sizes of Grinding Wheels, and for Shapes, Sizes and Identification of Mounted Wheels", and B74.13, "Markings for Identifying Grinding Wheels and Other Bonded Abrasives." The drive motor shall comply with National Electrical Manufacturers Association (NEMA) standard MG 1, "Motors and Generators." The feed mechanism shall be manually controlled and operated.

3.4 Accessories. Accessories such as automatic or semi-automatic feed mechanisms, wheel-truing devices, coolant systems, and work lights are not required unless otherwise specified in the acquisition order (see 7.4).

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3.5 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified in the acquisition order (see 7.4), the nameplate shall contain:

a. Nomenclature and classification (see table I)
b. Manufacturer's model designation
c. Electrical power requirements (voltage, full-load current, phase, frequency)
d. Short-circuit/over-current ratings
e. Manufacturer's name
f. Contract or order number
g. National stock number
h. Date of manufacture

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The grinding machines supplied shall meet the salient characteristics requirements of this CID; conform to the manufacturer's drawings, specifications, standards, and quality assurance practices; and be the same product as that which is offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. The drill bit grinding machines shall be preserved, packed, and marked as specified in the acquisition contract or order (see 7.4).

7. NOTES

7.1 Sources of documents

7.1.1 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002.

7.1.2 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.3 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

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7.2 National stock numbers (NSNs). The NSNs shown below are assigned for the listed machine types and sizes as specified in this document. Other NSNs may also apply.

Type	Size	Assigned NSN
I	4	3415-00-223-6567
		3415-01-347-2905
II	1	3415-00-242-5788
	2	3415-00-517-1050
		3415-00-242-5789

7.3 Sources of supply. The manufacturers listed below supply grinding machines which meet the salient characteristics requirements of this CID. Competition is not limited to the listed firms.

TDG Machine Products, Incorporated
PO Box 2029
Merrimack, NH 03054
(800) 811-0394

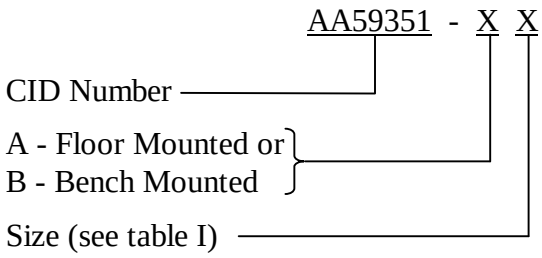
Service Precision Grinding
3204 State Highway 94
Franklin, NJ 07416
(973) 827-2271

RUSH Machinery, Incorporated
Route 364
Rushville, NY 14544
(716) 554-3070

7.4 Ordering data. Acquisition documents must specify:

a. Title, number, and date of this document
b. Type and size classification of grinding machine required (see table I)
c. Accessories required, if any (see 3.4)
d. Additional nameplate information, if any (see 3.5)
e. Packaging requirements (see 6.1)

7.5 Part identification number (PIN). The part identification numbering method shown below is intended for government purposes only and is not a requirement for any contractor.



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MILITARY INTERESTS:

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Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - AV
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CIVIL COORDINATING AGENCY:
GSA - 6FET

Preparing activity:
DLA - GS
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